

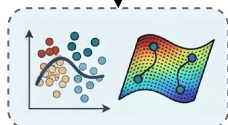
Certified Synthetic Data

Age	Education	Occupation	Income
35	Master's	Engineer	\$90k
...	...	...	...
...	...	...	...

Tabular Polygraph (HIF)

Age	Education	Occupation	Income
17	High School	Student	\$500
38	Master's	Engineer	\$125k
10	Primary	student	\$0
74	Bachelor's	Retired	\$28k

Real Tabular Data



Data Manifold

Mine Rules

Train Sentinels

Fit SVD / GBR

NIC Violation

Evaluate Record

Synthetic Tabular Data

Age	Education	Occupation	Income
17	Bachelor's	student	\$850k
5	High School	student	\$0
35	Master's	Engineer	\$90k
22	Medical degree	Brain Surgeon	\$75k

Rule Violation

LSE Violation



Logic rules
IF (A, B) THEN (C)
IF (Age < 18)
IF (Region='USA')

Structural Logic Constraints



Hub Discovery (S(h))



Sentinels $P(x_h | x_{lh})$

Logical Sentinel Ensemble



Local data point comparisons



Smooth embedding manifolds

Neighbor-Invariant Continuity

Structural Penalty

Categorical Penalty

Continuous Penalty

P_{rule}

P_{cat}

P_{cont}



Multiplicative Aggregation

Yes ✓

No ✗



Hallucinations

HIF Score > Threshold?